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The Premier Conference & Exhibition on
Computer Graphics & Interactive Techniques

NeuBase: Spline Surfaces with Neural Basis Functions

Anshul Mendiratta, Lei Yang, Xin Li,
John Keyser, Scott Schaefer, Wenping Wang

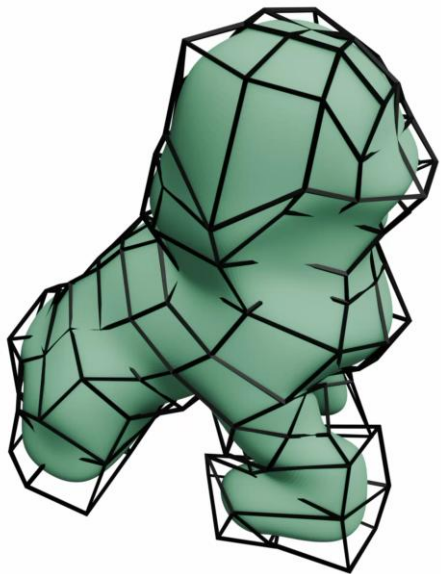


2026



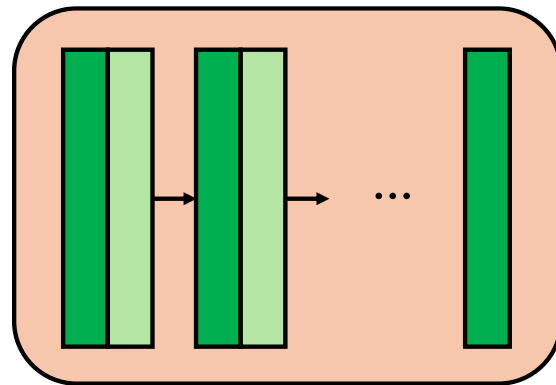
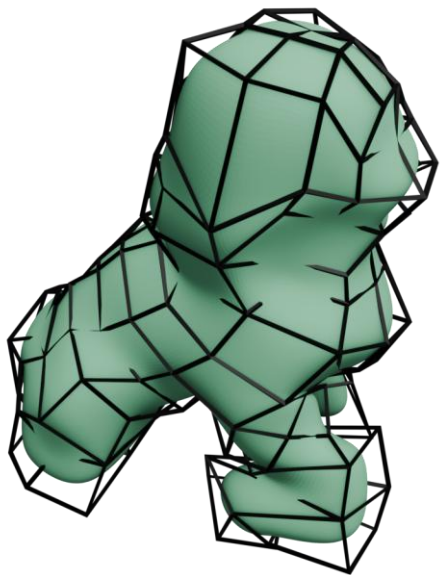
What is NeuBase?

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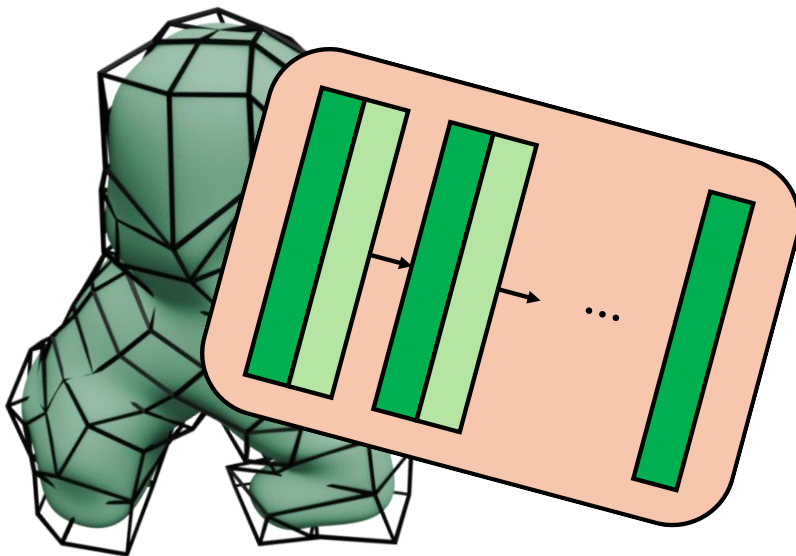
What is NeuBase?

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What is NeuBase?

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Catmull-Clark Surfaces

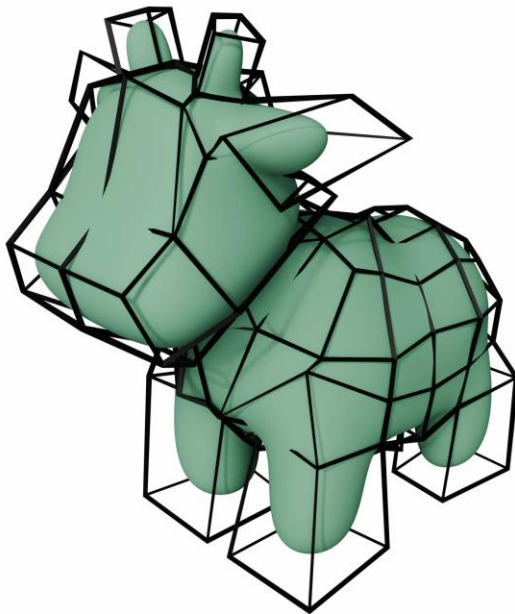


Properties of
Catmull-Clark
surfaces

Arbitrary topology

Smooth

Editable



Catmull-Clark Surfaces



**Limitations of
Catmull-Clark
surfaces**

Control mesh density ↑

Storage cost ↑

Computational cost ↑

NeuBase Surfaces

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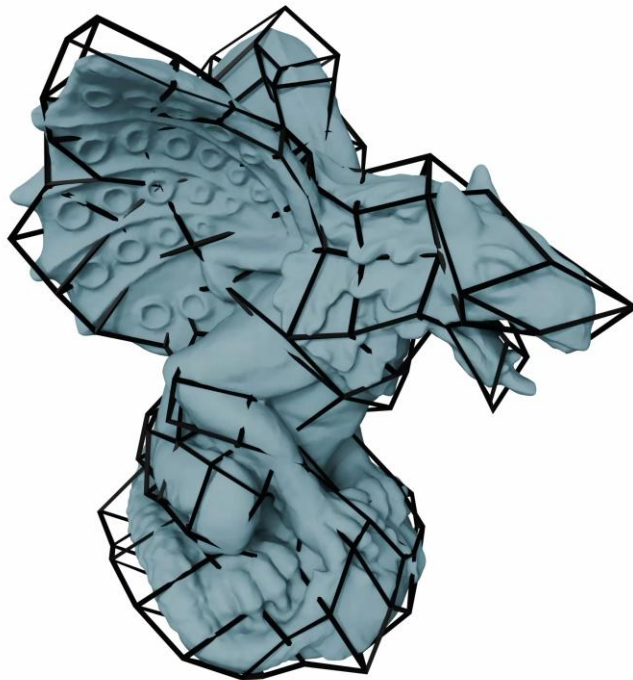


Advantages of
NeuBase surfaces

Arbitrary topology

Smooth

Editable

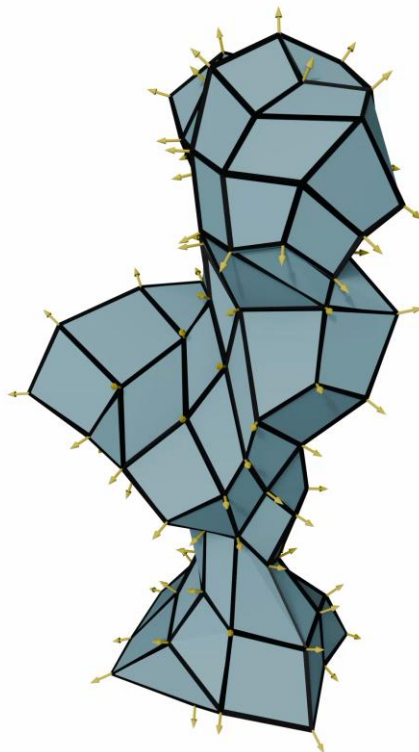


Control mesh density ↓

Storage cost ↓

Computational cost ↓

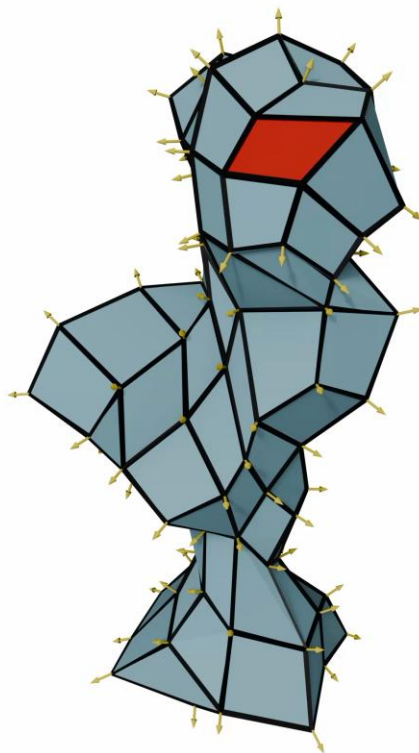
NeuBase Framework



We use Quadriflow to
extract the quad control
mesh

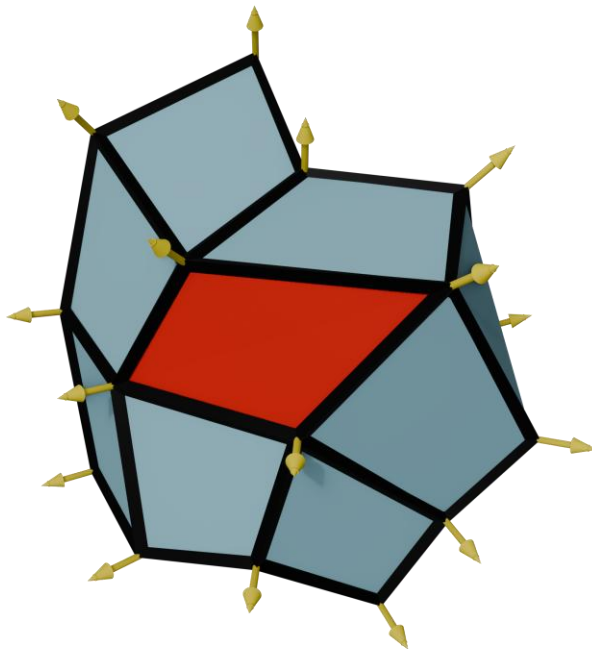
Jingwei Huang, Yichao Zhou,
Matthias Nießner, Jonathan
Shewchuk, Leonidas Guibas

NeuBase Framework

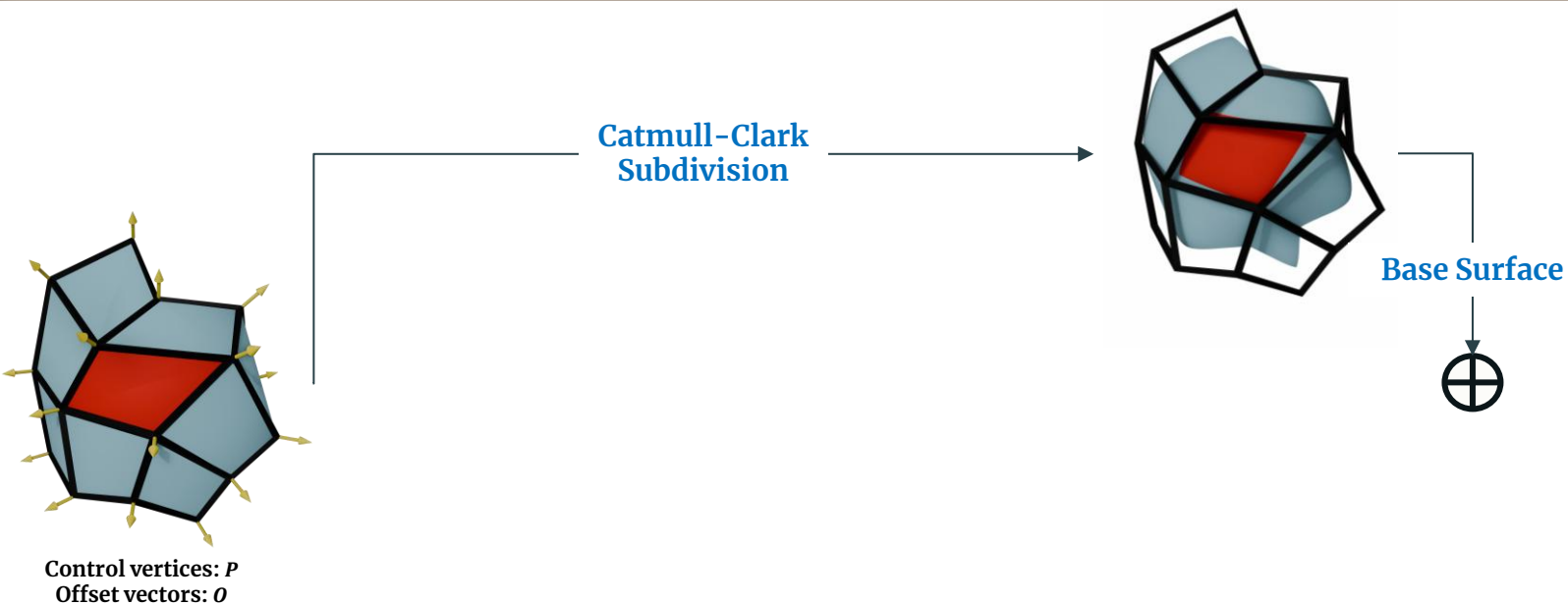


NeuBase Framework

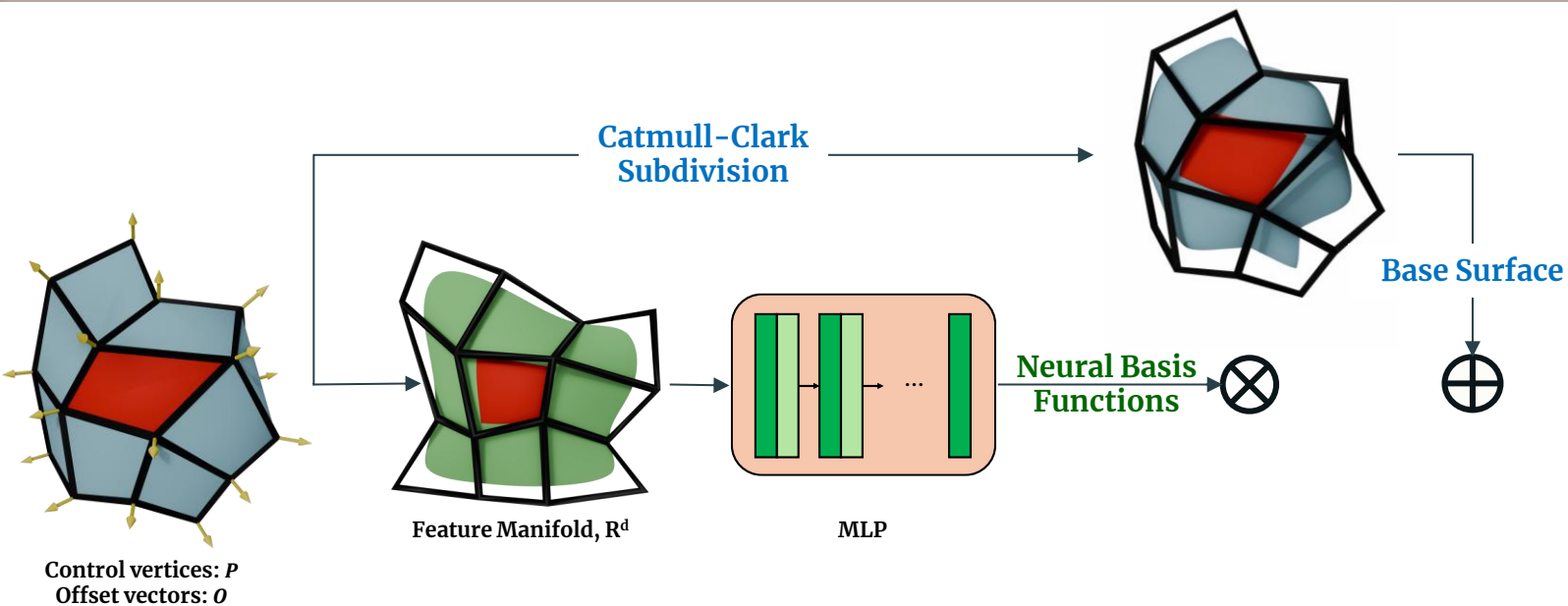
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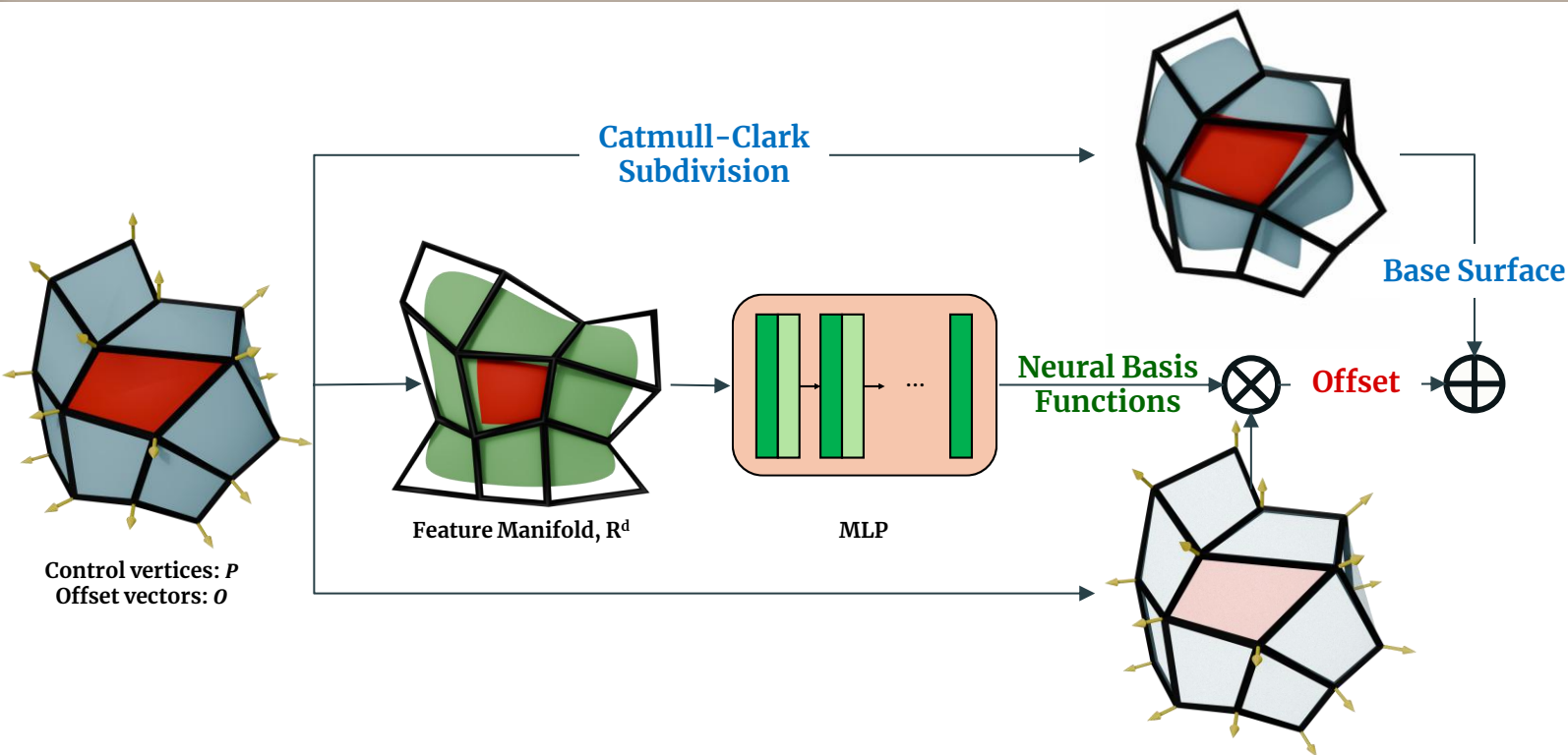
NeuBase Framework



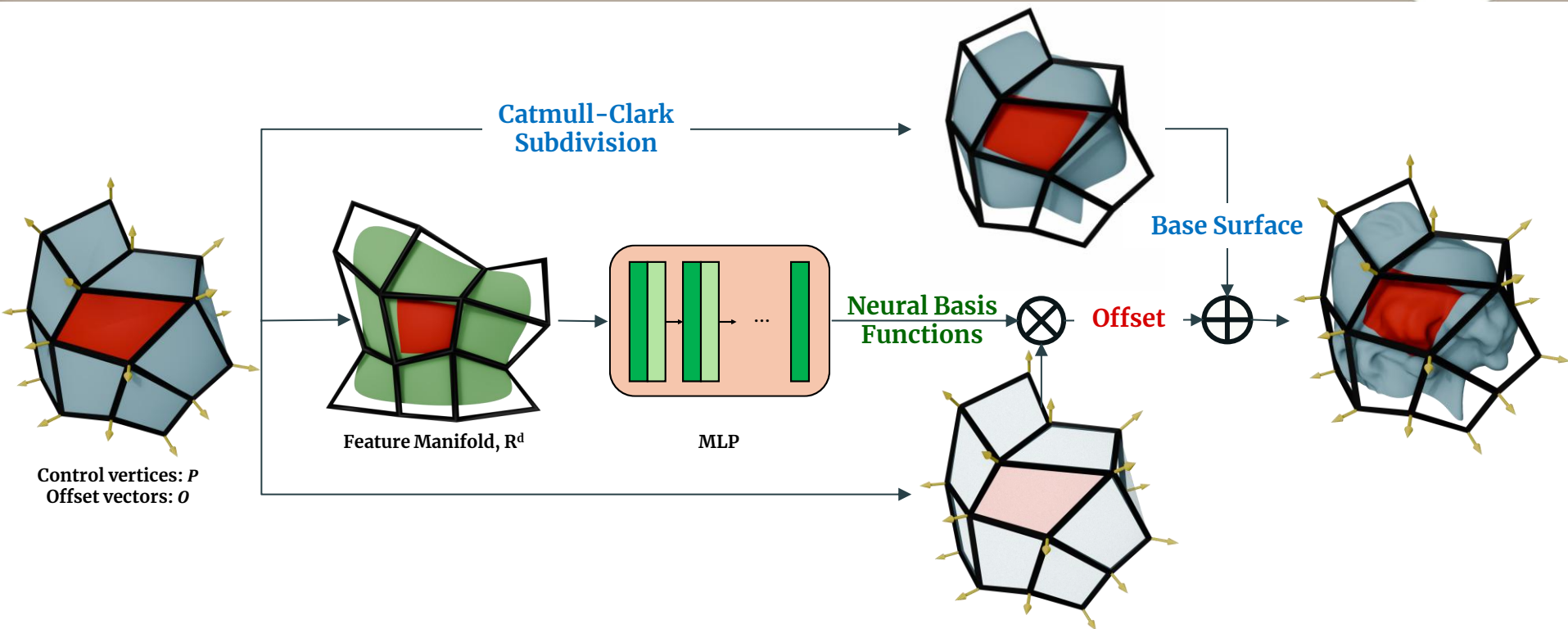
NeuBase Framework



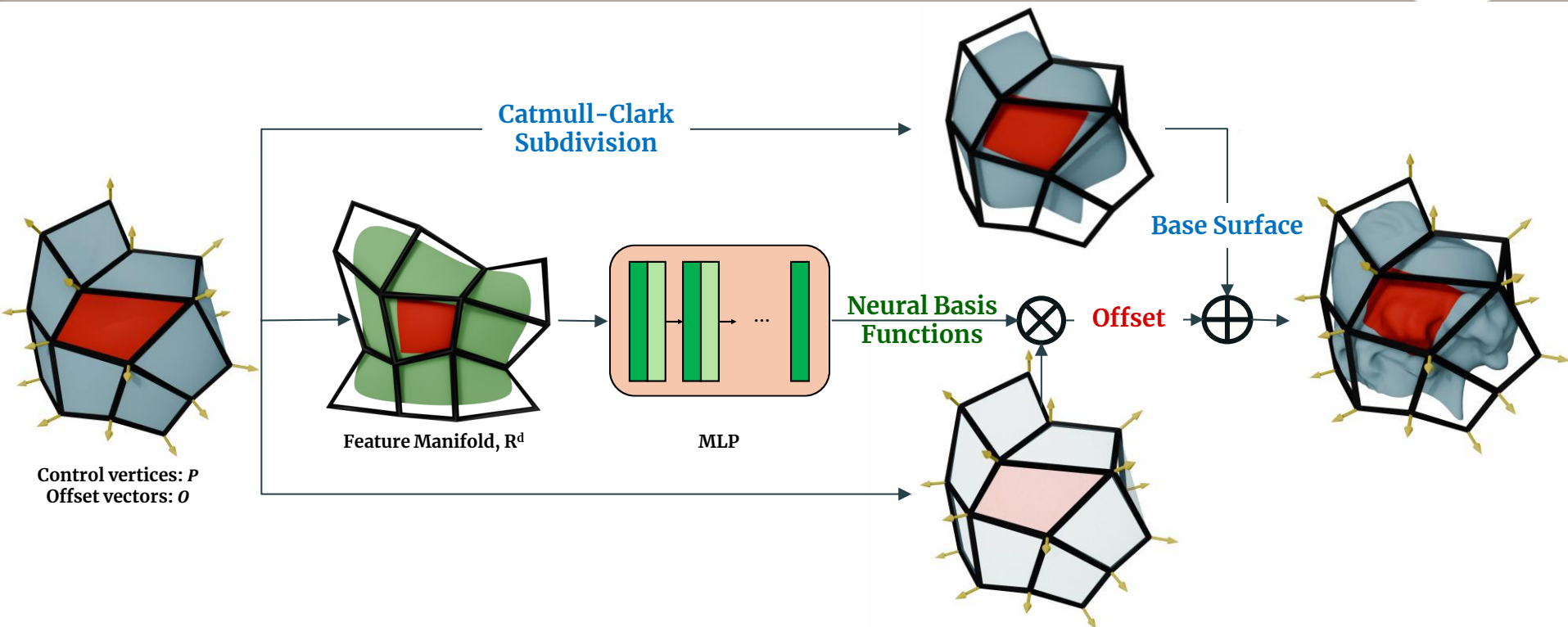
NeuBase Framework



NeuBase Framework

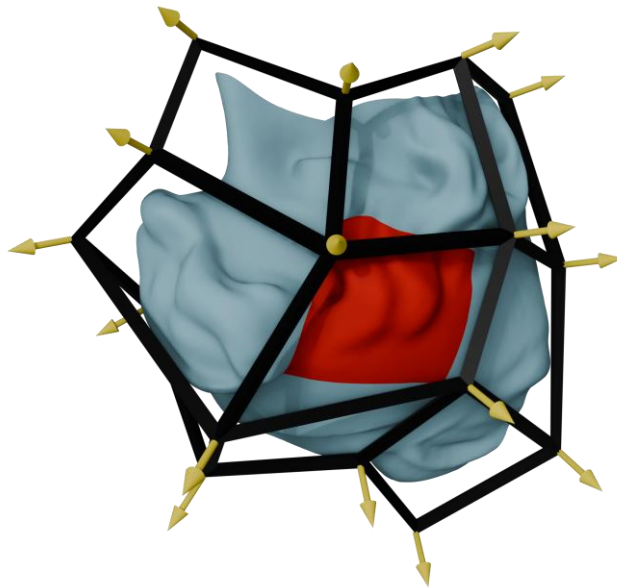


NeuBase Framework



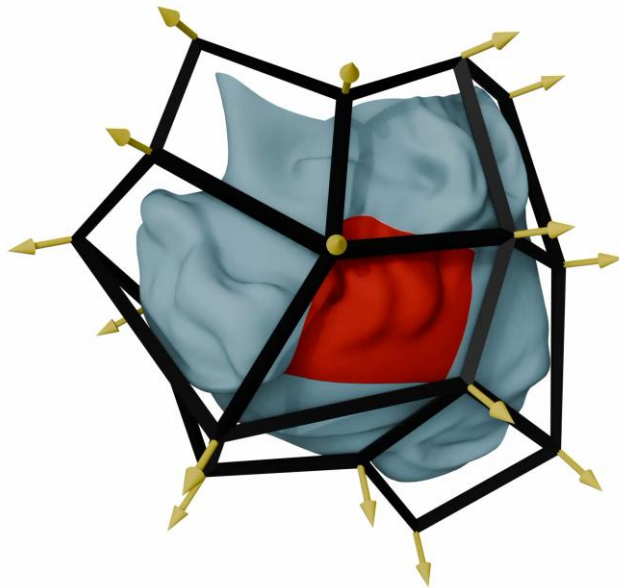
NeuBase Framework

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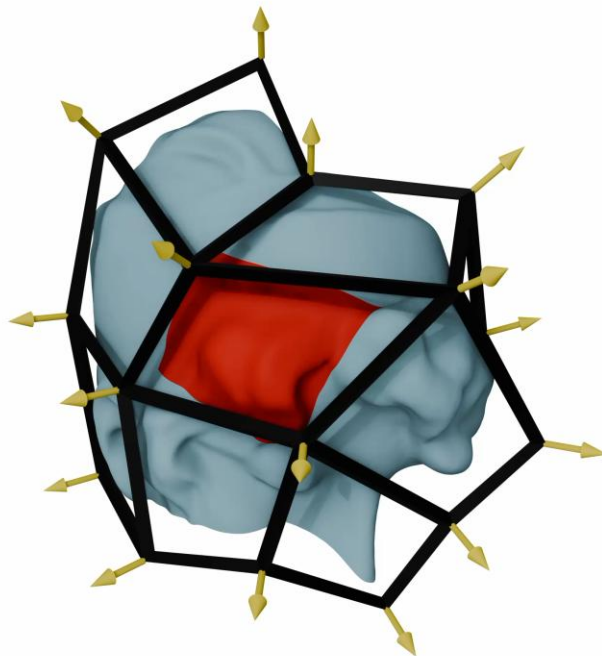


NeuBase Framework

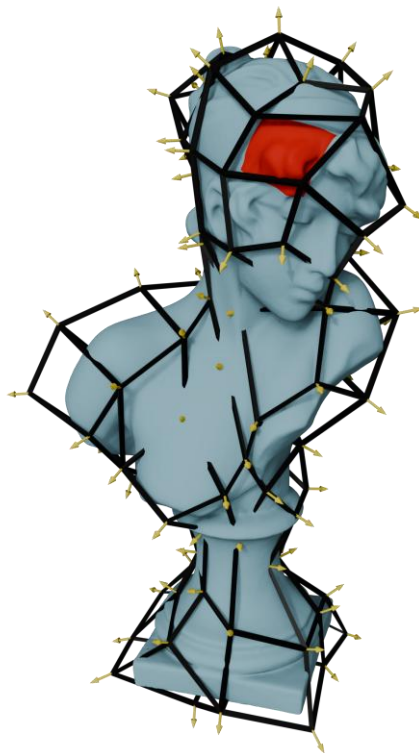
SIGGRAPH



NeuBase Framework



NeuBase Framework

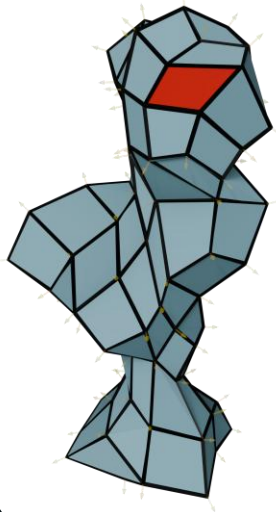


NeuBase Framework



Base Surface

Control vertices: P

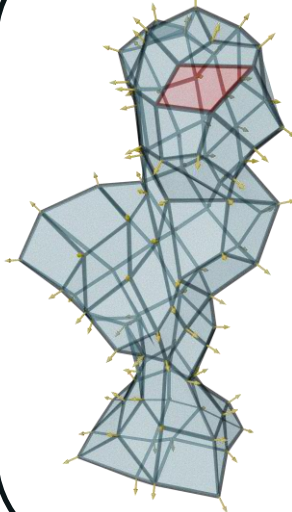


Catmull-Clark
Basis Functions: B

$$S_B(u, v) = \sum_i B_i(u, v) P_i$$

Offset

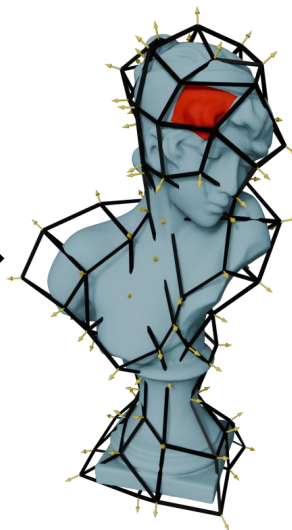
Offset vectors: O



Neural Basis
Functions: N

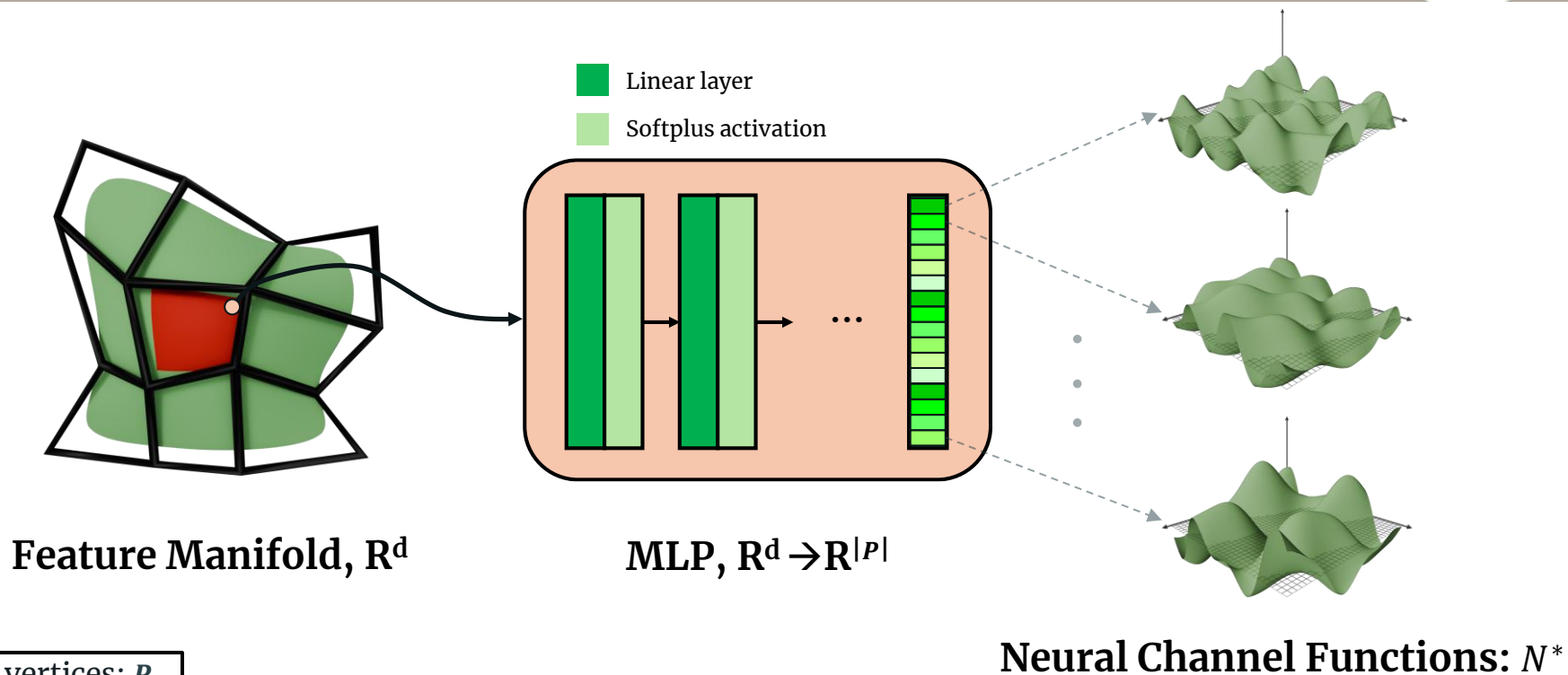
$$S_O(u, v) = \sum_i N_i(u, v) O_i$$

+



$$S(u, v) = S_B(u, v) + S_O(u, v)$$

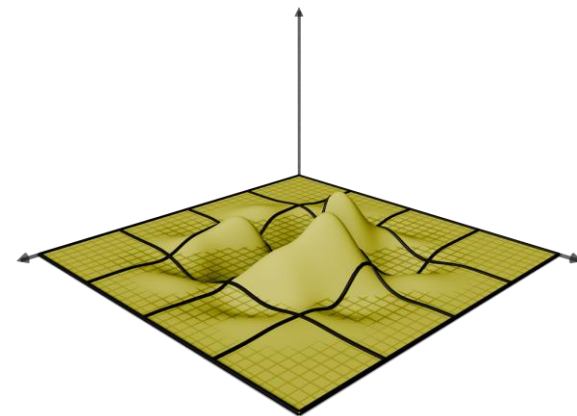
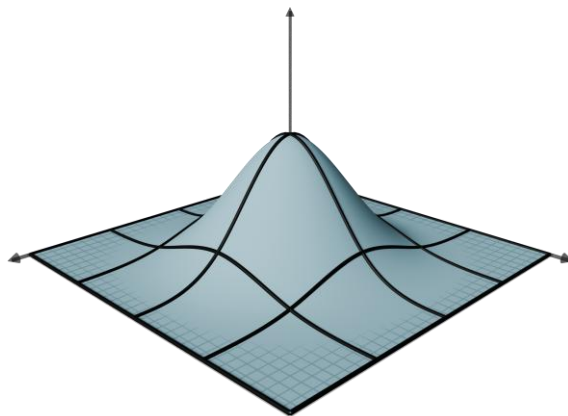
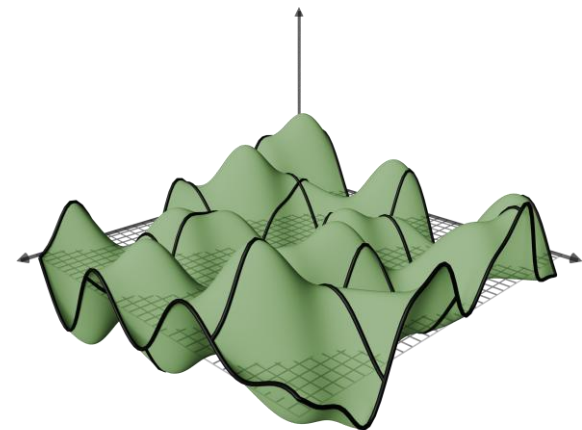
Neural Basis Function Design



Neural Basis Function Design



This step enforces Neural Basis Functions to have the same **local support** and **smoothness** as Catmull-Clark basis functions



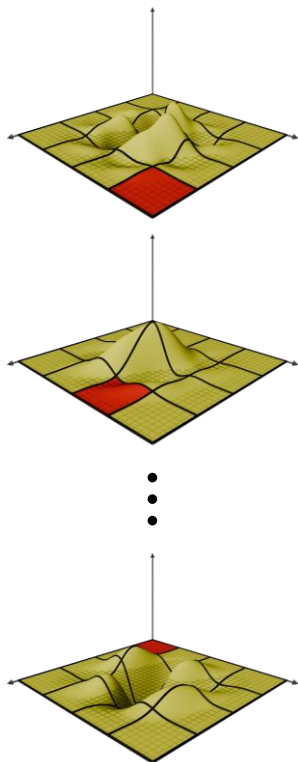
Neural Channel Function: N_i^*

Catmull-Clark Basis Function: B_i

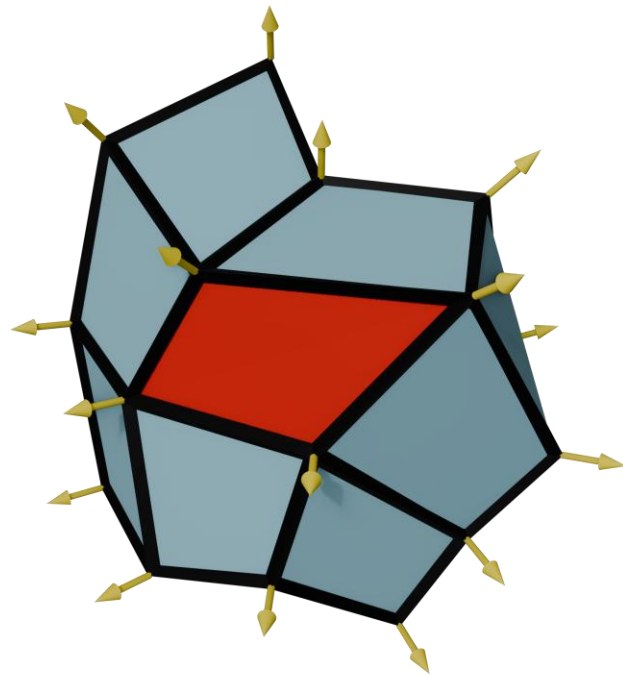
Neural Basis Function: N_i

Our solution

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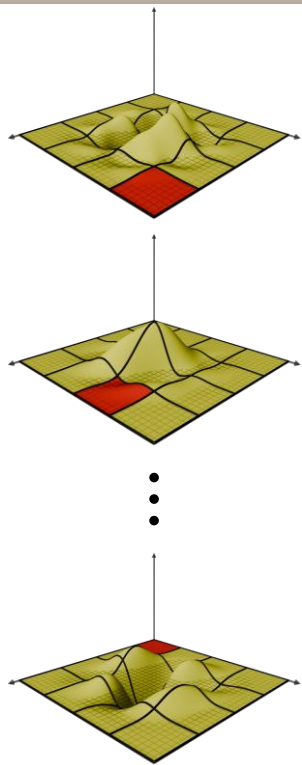
$\odot$



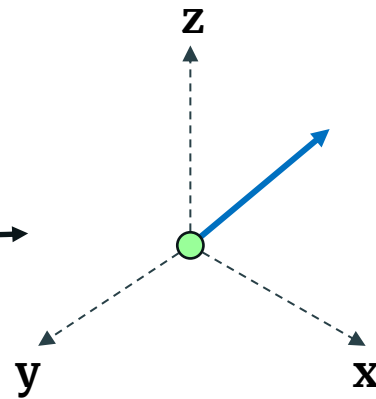
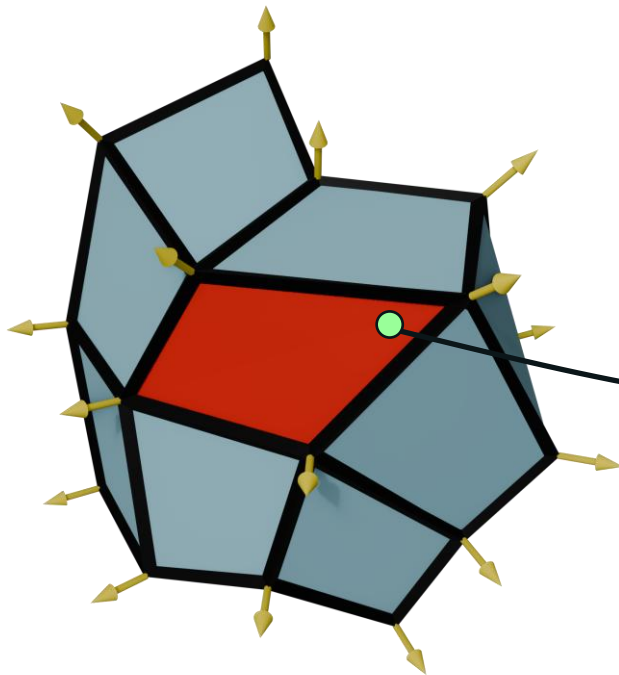
$\odot$ Linear combination

Our Solution

SIGGRAPH



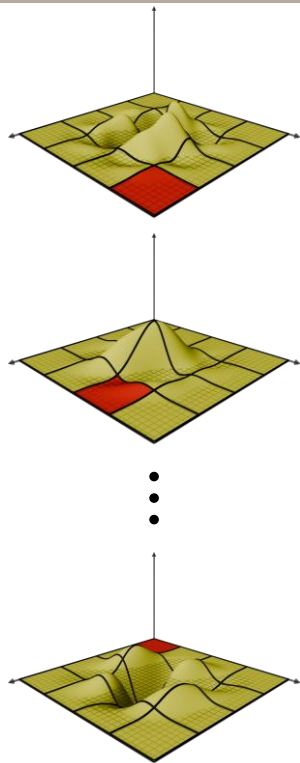
○



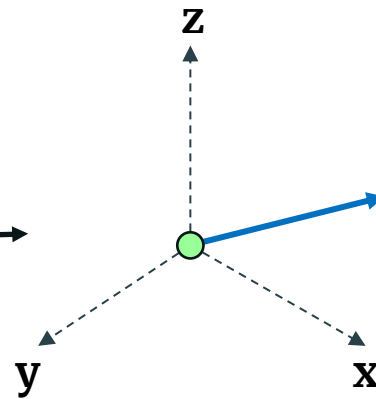
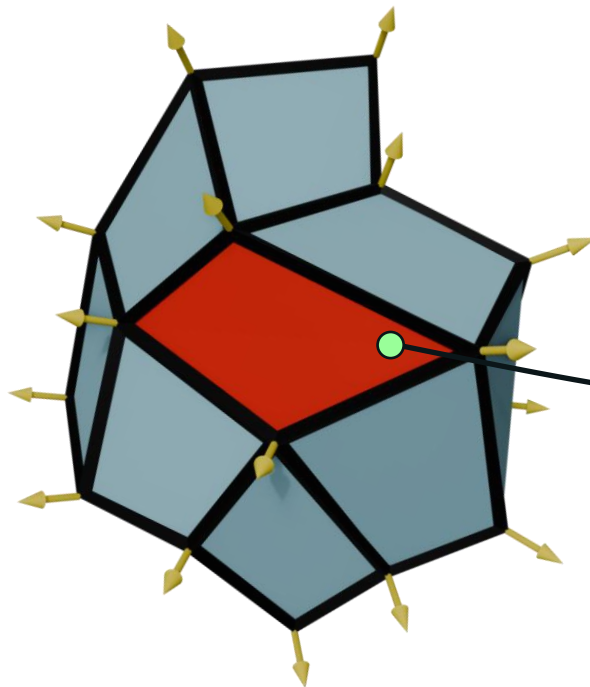
○ Linear combination

Our Solution

SIGGRAPH



○



○ Linear combination

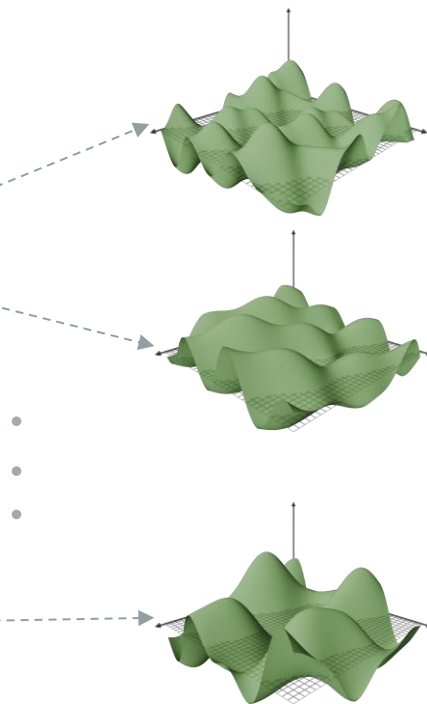
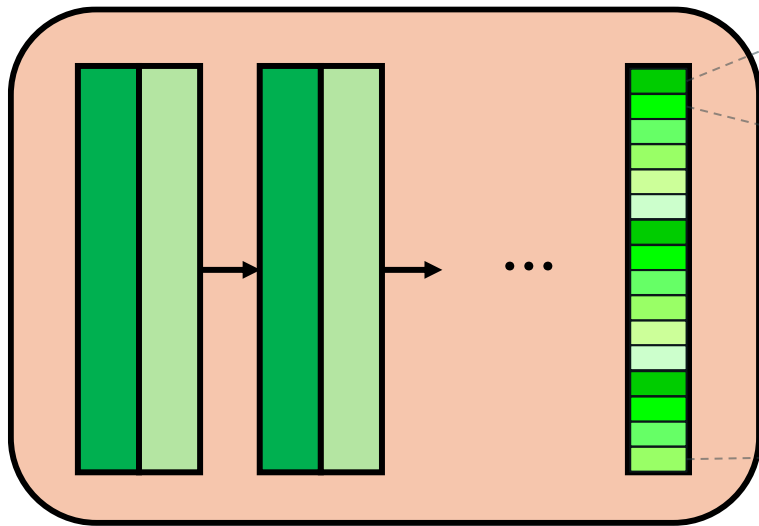


Network Design

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The size of the network **increases**
with the number of control vertices, $|P|$



One-to-one correspondence between output nodes and P

Network Design

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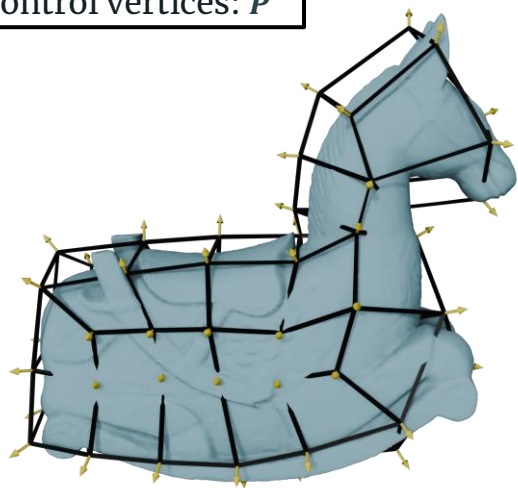


Rate of increase in storage size w.r.t. P

Control vertices: P

2x

7x



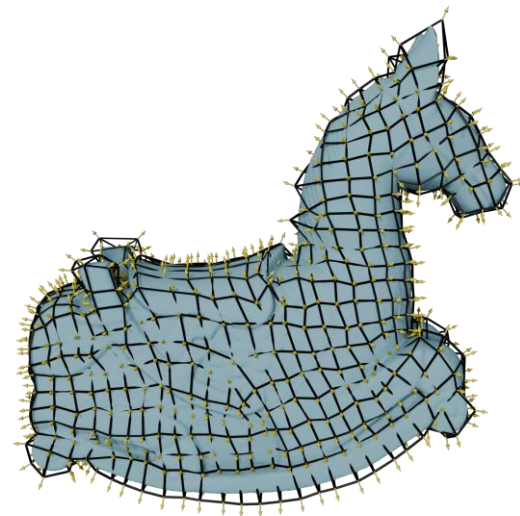
$|P| = 63$

Storage = 185KB



$|P| = 250$

Storage = 375KB

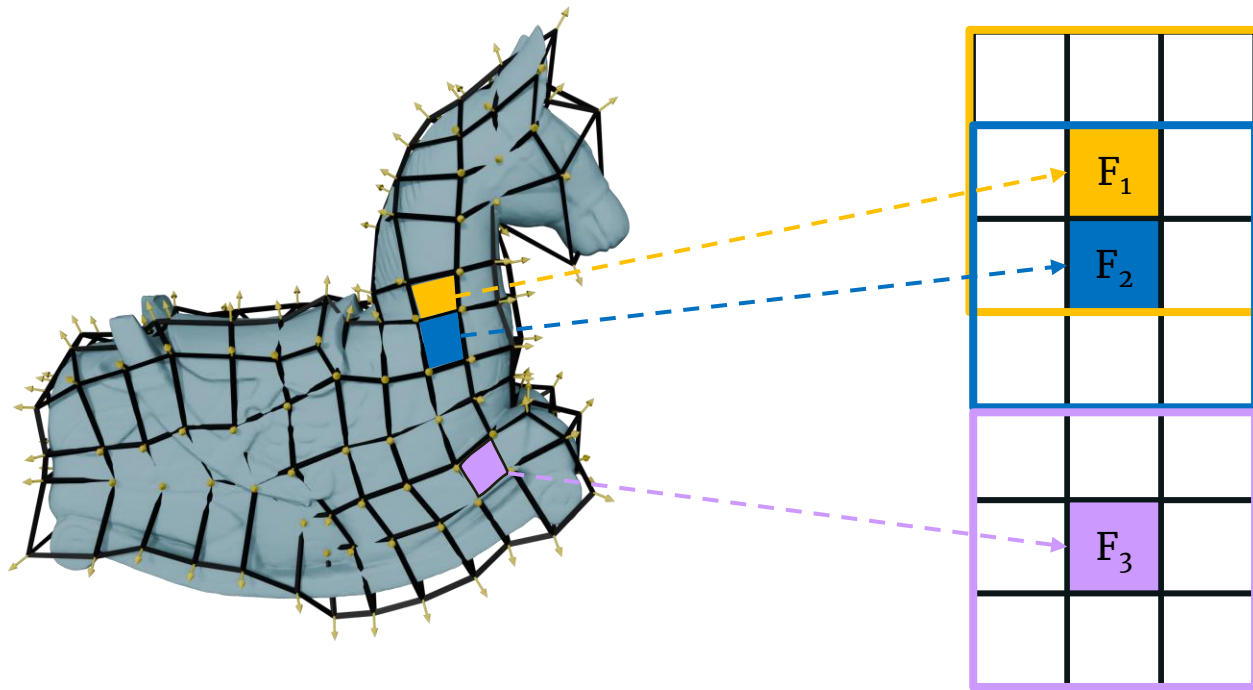


$|P| = 1000$

Storage = 1330KB

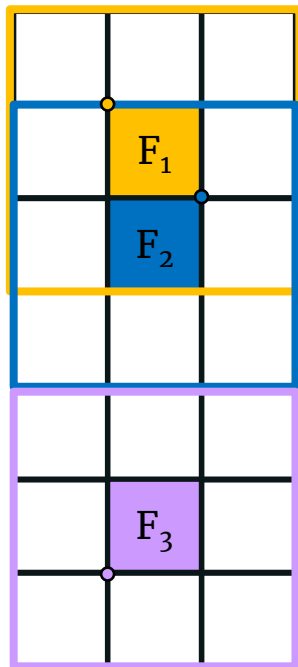
Network Design

SIGGRAPH



Network Design

SIGGRAPH



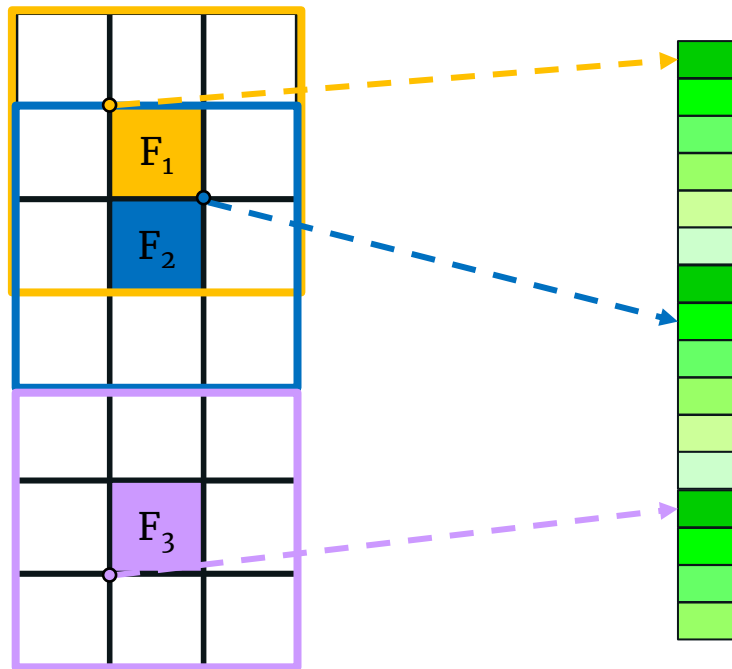
Control vertices: P



Output layer, $R^{|P|}$

Network Design

SIGGRAPH



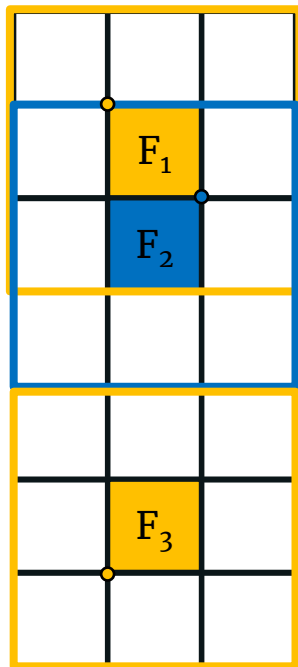
Each control vertex is assigned a **unique** channel irrespective of the patch it belongs to

Control vertices: P

Output layer, $R^{|P|}$

Our Solution

SIGGRAPH



K: Number of channels

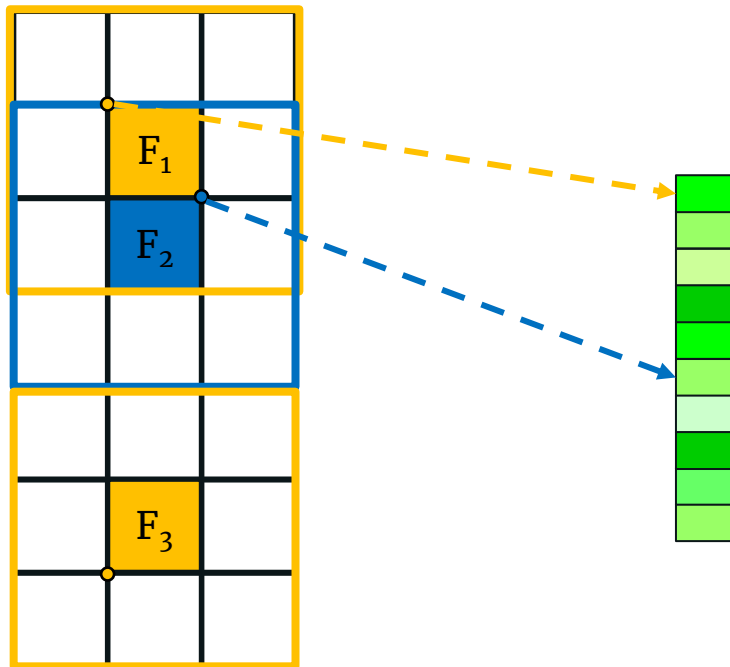
Control vertices: P

Output layer, $R^{|\mathbf{K}|}$

$K \ll |P|$

Our Solution

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Control vertices
influencing the same
patch are assigned
different channels

K: Number of channels

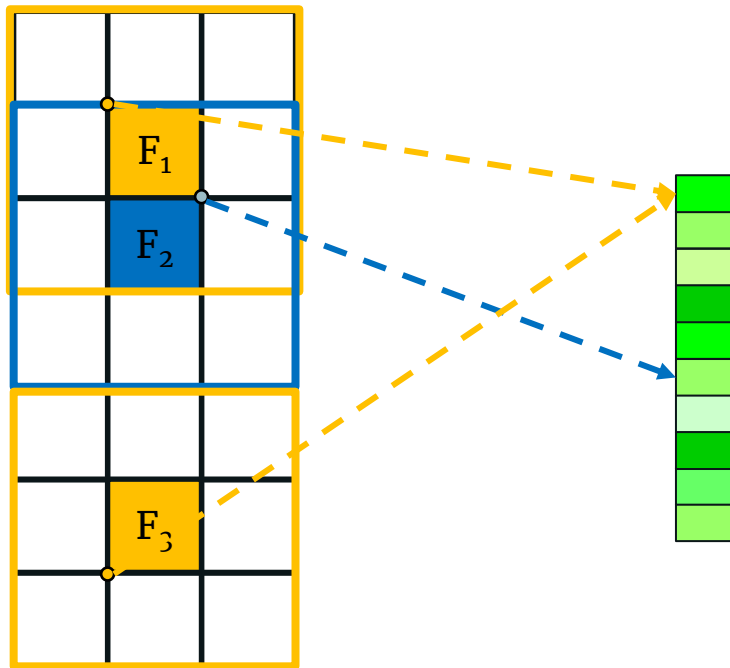
Control vertices: P

Output layer, R^K

$K \ll |P|$

Our Solution

SIGGRAPH



Control vertices
influencing the same
patch are assigned
different channels

Control vertices
influencing different
patches can share the
same channel

K: Number of channels

Control vertices: P

Output layer, $R^{|K|}$

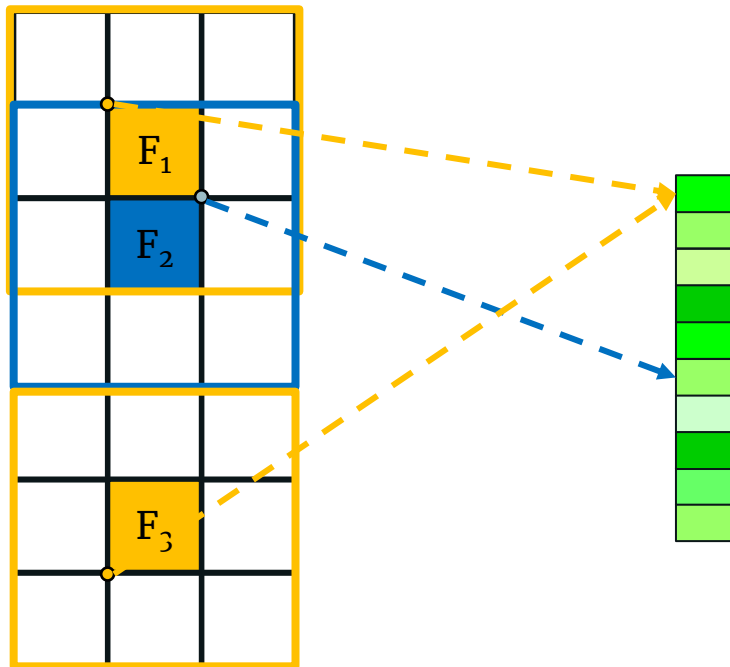
$K \ll |P|$

Our Solution

SIGGRAPH



This mapping can be obtained by treating this as an **equitable graph coloring** problem



Control vertices influencing the same patch are assigned **different** channels

Control vertices influencing different patches can share the **same** channel

K : Number of channels

Control vertices: P

Output layer, $R^{|K|}$

$K \ll |P|$

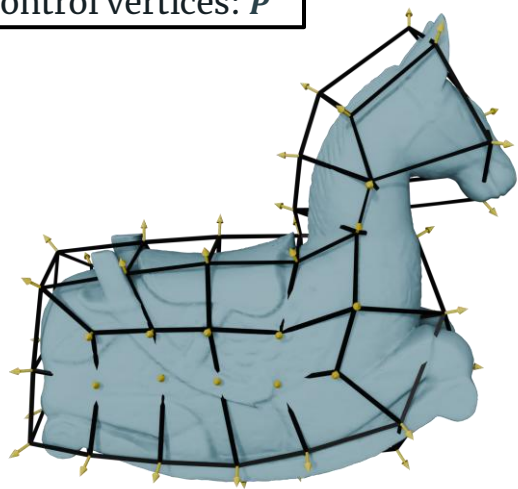
Our Solution

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Improved storage size with our design

Control vertices: P



$|P| = 63$

Original storage = 185KB

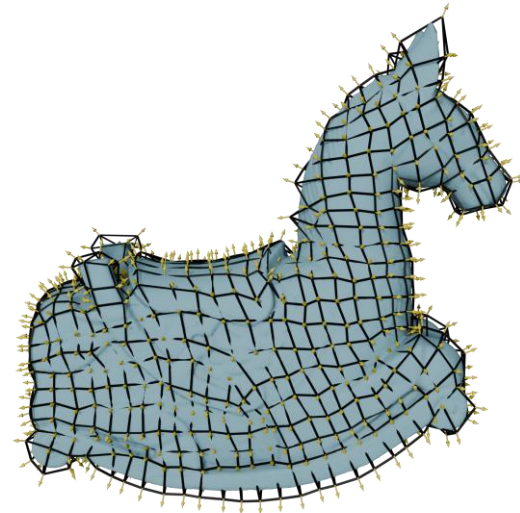
New storage = 138KB



$|P| = 250$

Original storage = 375KB

New storage = 153KB



$|P| = 1000$

Original storage = 1330KB

New storage = 230KB

Effect of K on Reconstruction Accuracy



$K = 1$

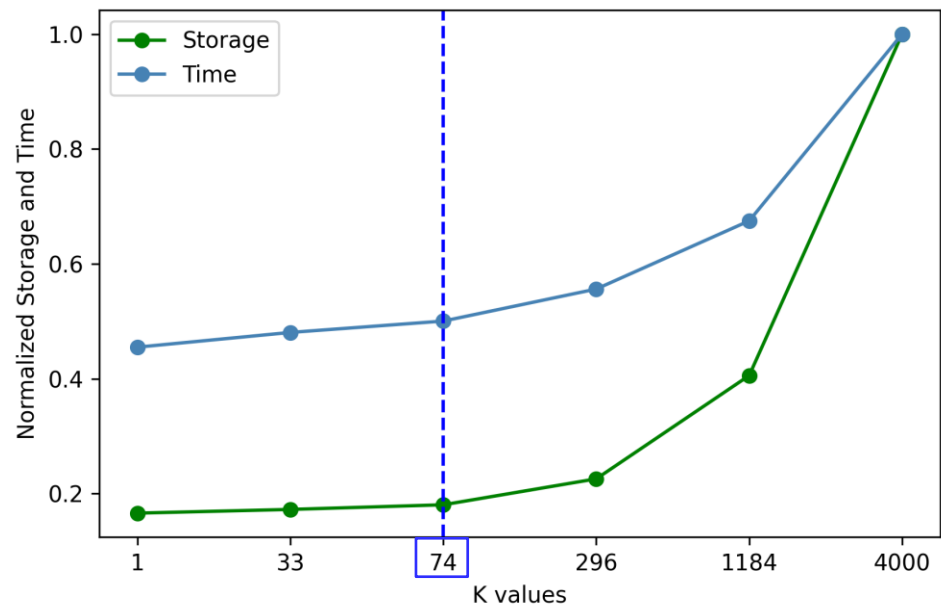
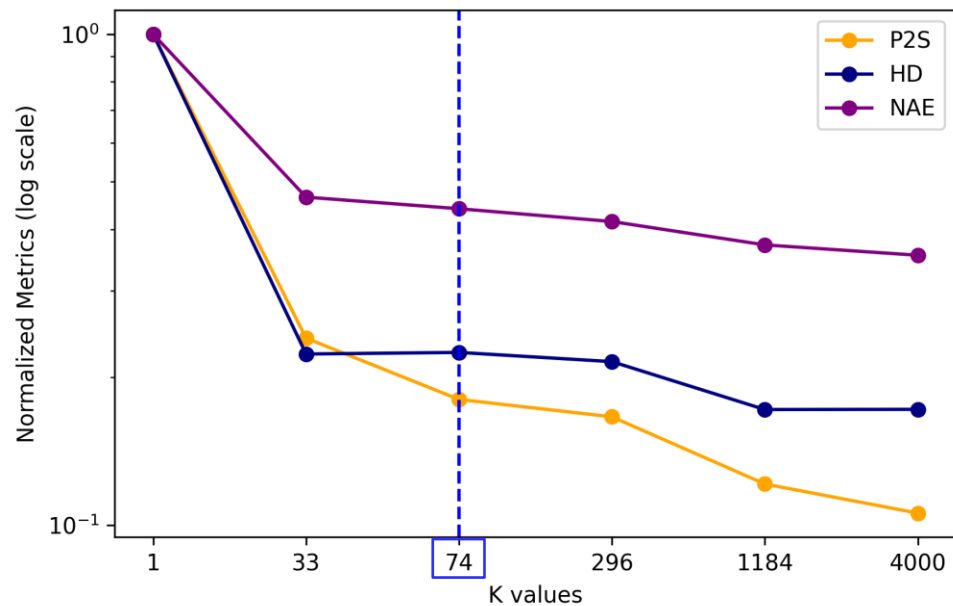
$K = 74$

$K = 296$

$K = 1184$

$K = |P| = 4000$

Effect of K on Reconstruction Accuracy



Comparison with Other Methods



P2S: Point to surface error

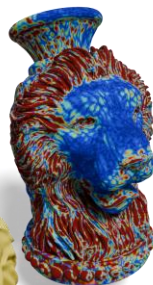


Groundtruth



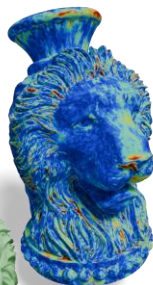
Catmull-Clark

P2S: 22.27×10^{-4}
Storage: 175 KB



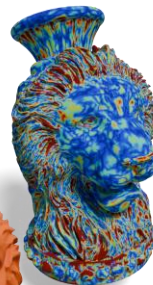
NGF

P2S: 7.06×10^{-4}
Storage: 261 KB



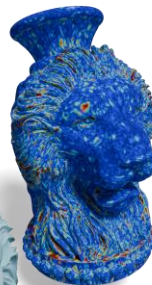
NeuPPS

P2S: 13.63×10^{-4}
Storage: 167 KB



NeuBase (Ours)

P2S: 6.22×10^{-4}
Storage: 167 KB



Comparison with Other Methods



P2S: Point to surface error



Groundtruth

Catmull-Clark

NGF

NeuPPS

NeuBase (Ours)

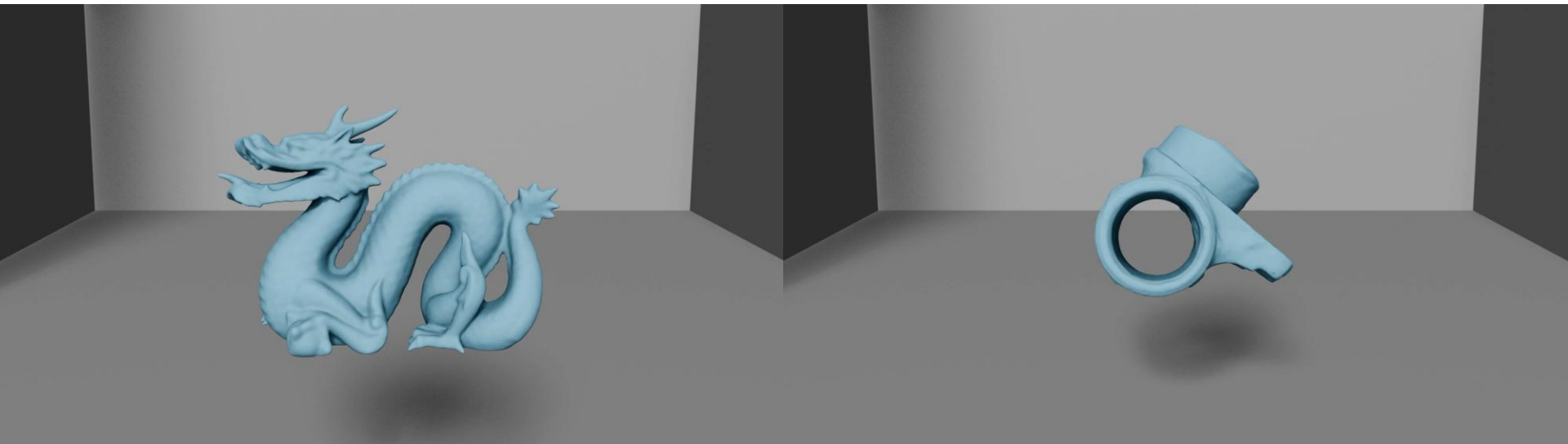
P2S: 32.60×10^{-4}
Storage: 165 KB

P2S: 6.47×10^{-4}
Storage: 156 KB

P2S: 3.17×10^{-4}
Storage: 144 KB

P2S: 2.28×10^{-4}
Storage: 141 KB

Deformation Examples



Thank You